

Predictive Model Plan – Student Template

1. Model Logic (Generated with GenAI)

The predictive model is designed to forecast whether a customer is likely to become delinquent based on their financial and behavioral information. The system analyzes important variables such as income, credit utilization, missed payments, debt-to-income ratio, and credit score to estimate delinquency risk.

Simplified Pseudocode:

- *Load dataset*
- *Handle missing values*
- *Split data into training and testing sets*
- *Train Random Forest model*
- *Predict delinquency outcomes*
- *Evaluate model performance using:*
 - *Accuracy*
 - *Precision*
 - *Recall*
 - *F1-Score*

2. Justification for Model Choice

The Random Forest Classifier was selected because it performs well in classification problems involving complex financial behaviour patterns. Since credit delinquency is influenced by multiple interacting factors, Random Forest is suitable for capturing nonlinear relationships and improving prediction accuracy.

Reasons for Selecting This Model

- **Accuracy:**
Random Forest generally provides strong predictive performance and reduces overfitting through ensemble learning.
- **Transparency:**
Although more complex than logistic regression, feature importance analysis can still explain which variables most influence delinquency predictions.
- **Ease of Implementation:**
The model is widely supported in machine learning libraries such as Scikit-learn and can be implemented efficiently.
- **Financial Relevance:**
Financial risk prediction often involves complicated customer behavior patterns, making Random Forest suitable for identifying delinquency risk.

- **Business Suitability for Geldium:**

The model can help Geldium proactively identify high-risk customers and support better decision-making for collections and risk management.

3. Evaluation Strategy

The model's performance will be evaluated using multiple classification metrics to ensure reliable delinquency predictions.

Evaluation Metrics

- **Accuracy:** Measures the overall percentage of correct predictions.
- **Precision:** Evaluates how many predicted delinquent customers were actually delinquent.
- **Recall:** Measures how effectively the model identifies customers who are truly at risk.
- **F1-Score:** Balances precision and recall to provide a more reliable performance measure.
- **ROC-AUC Score:** Measures the model's ability to distinguish between delinquent and non-delinquent customers.

Interpretation of Metrics

- High recall is important because missing high-risk customers could lead to financial losses.
- Precision is also important to avoid incorrectly labelling low-risk customers as delinquent.
- The F1-score helps balance both concerns.

Bias Detection and Reduction

To reduce model bias:

- Missing values will be handled carefully using appropriate imputation methods.
- Model performance will be checked across different customer groups.
- Feature importance will be reviewed to ensure no unfair variables dominate predictions.

Ethical Considerations

When predicting customer financial behaviour:

- Customer data privacy must be protected.
- Predictions should support fair financial decisions and avoid discrimination.
- AI-generated predictions should assist human decision-making rather than fully replace it.